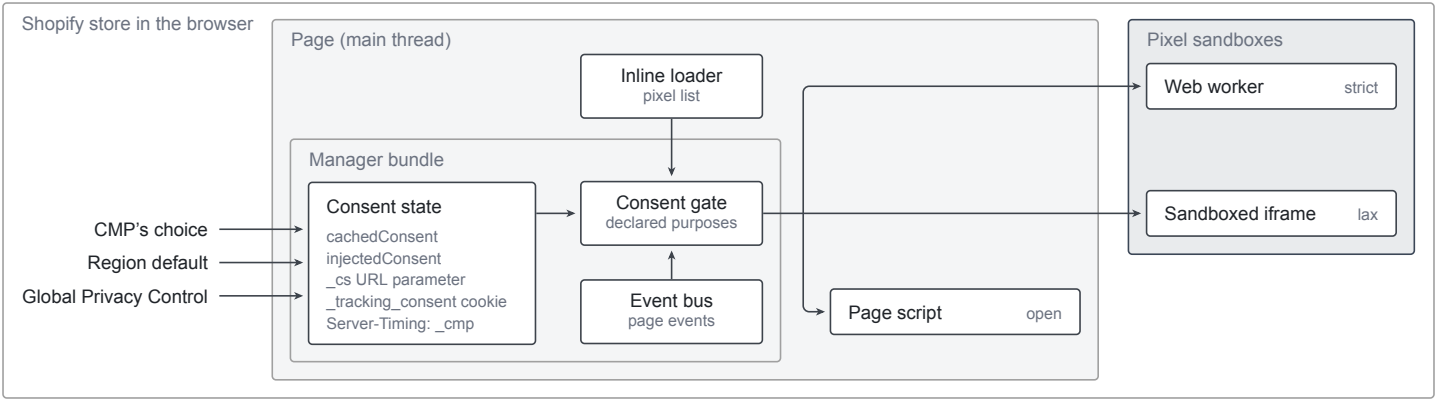
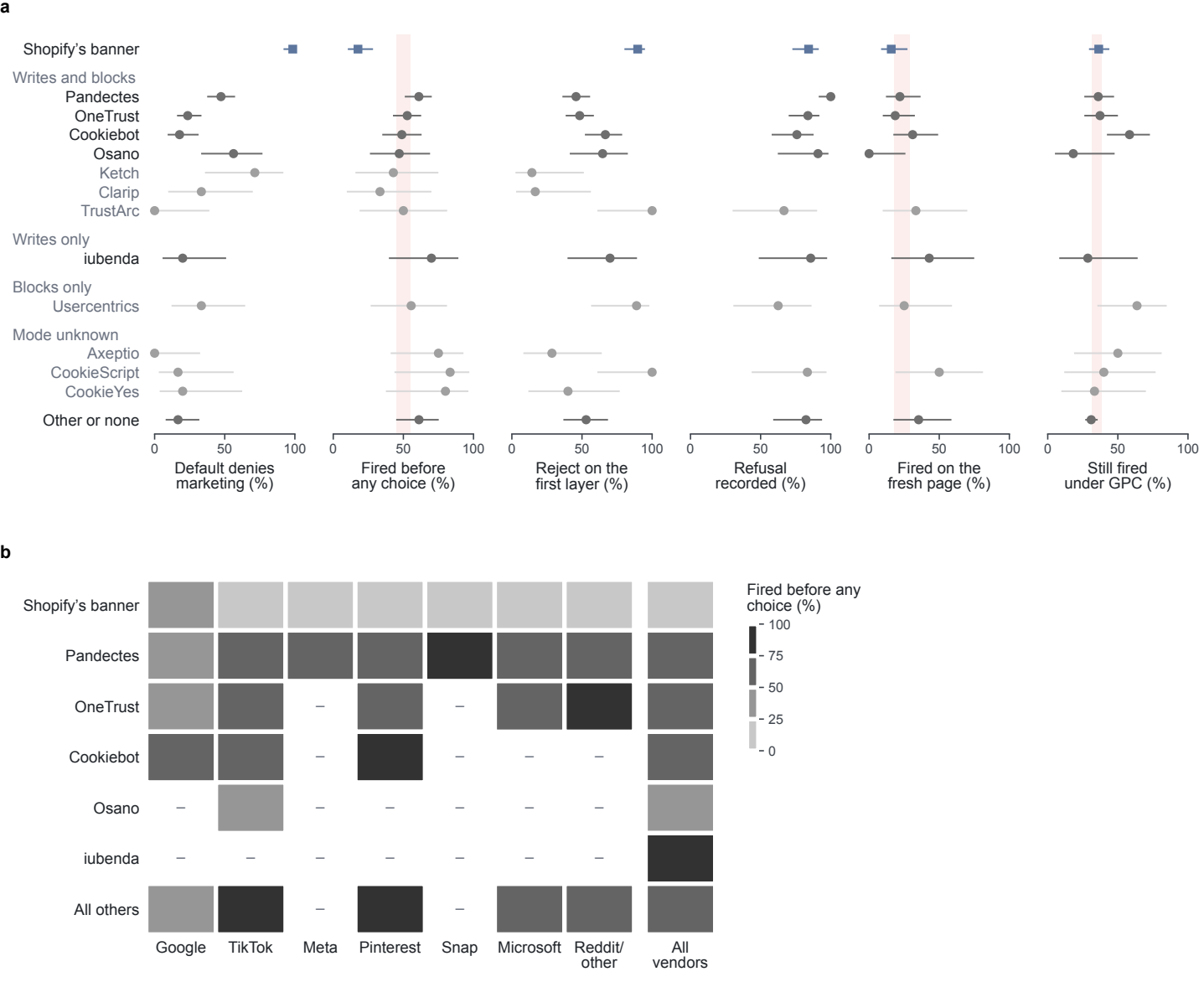
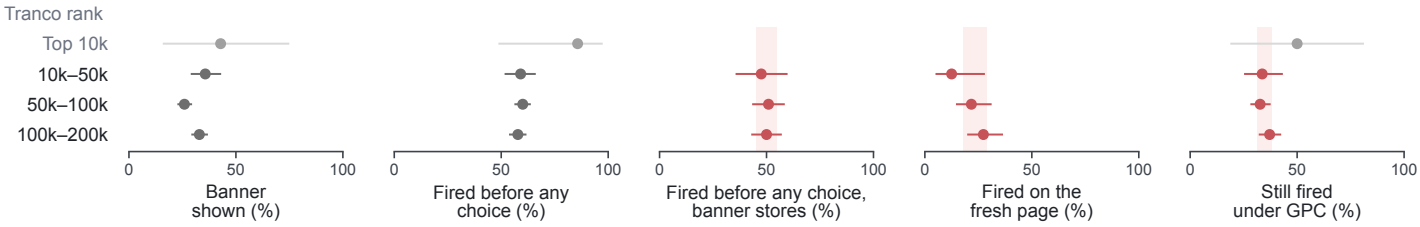
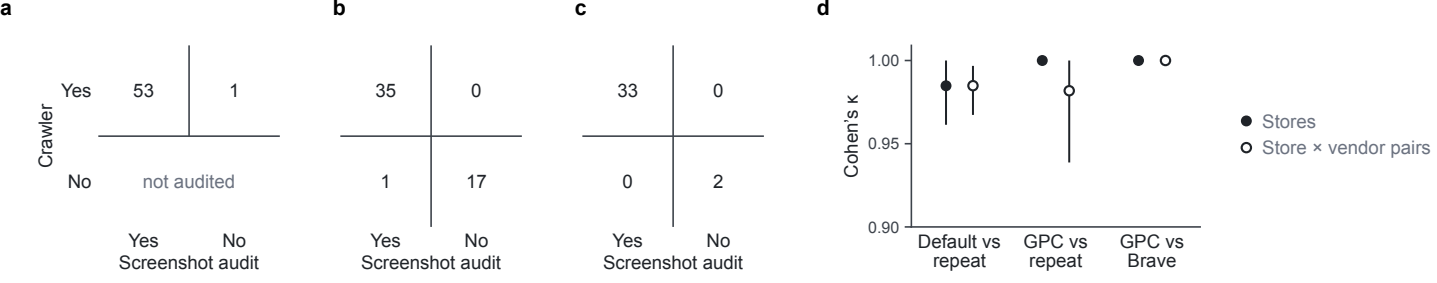
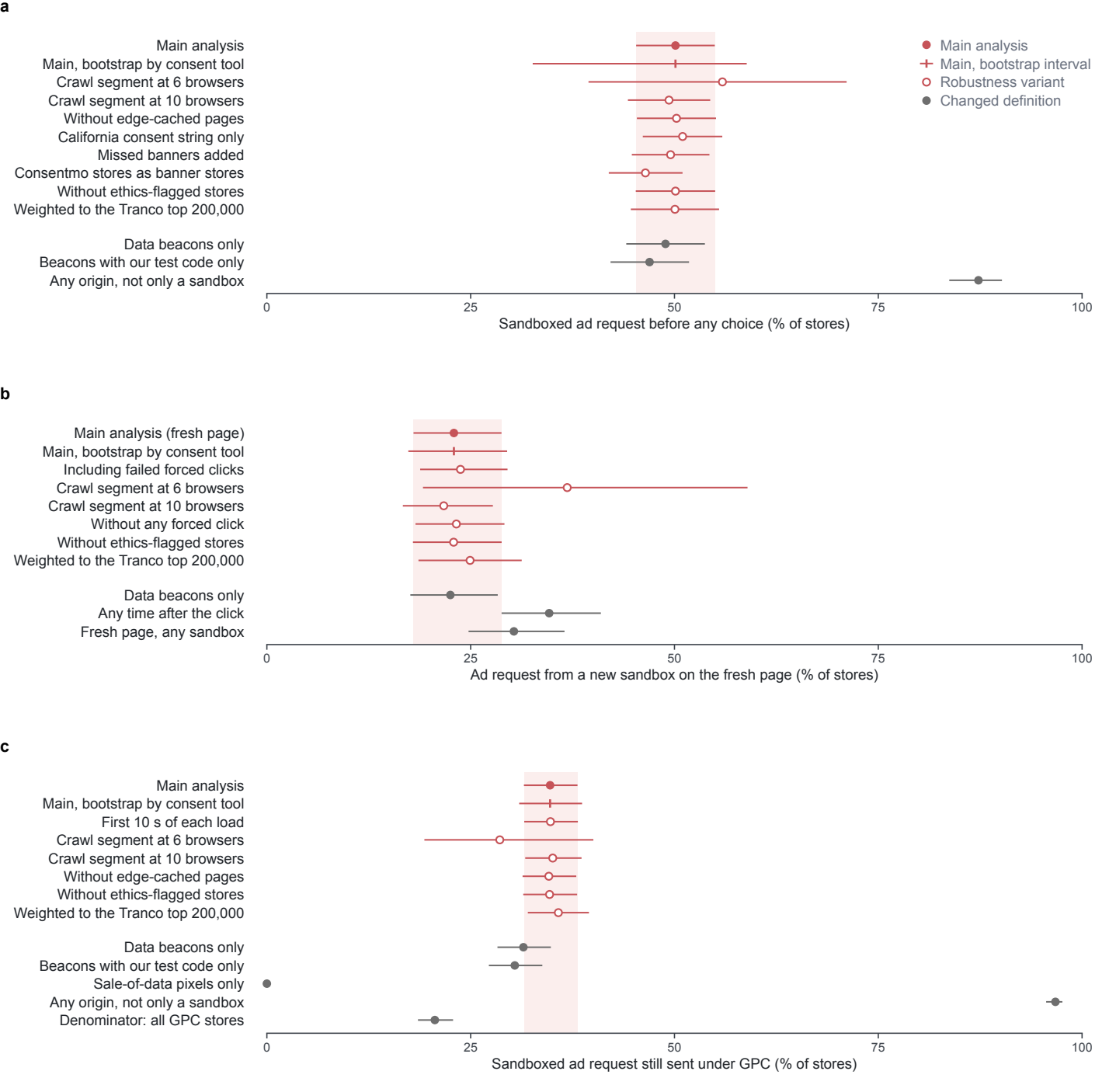
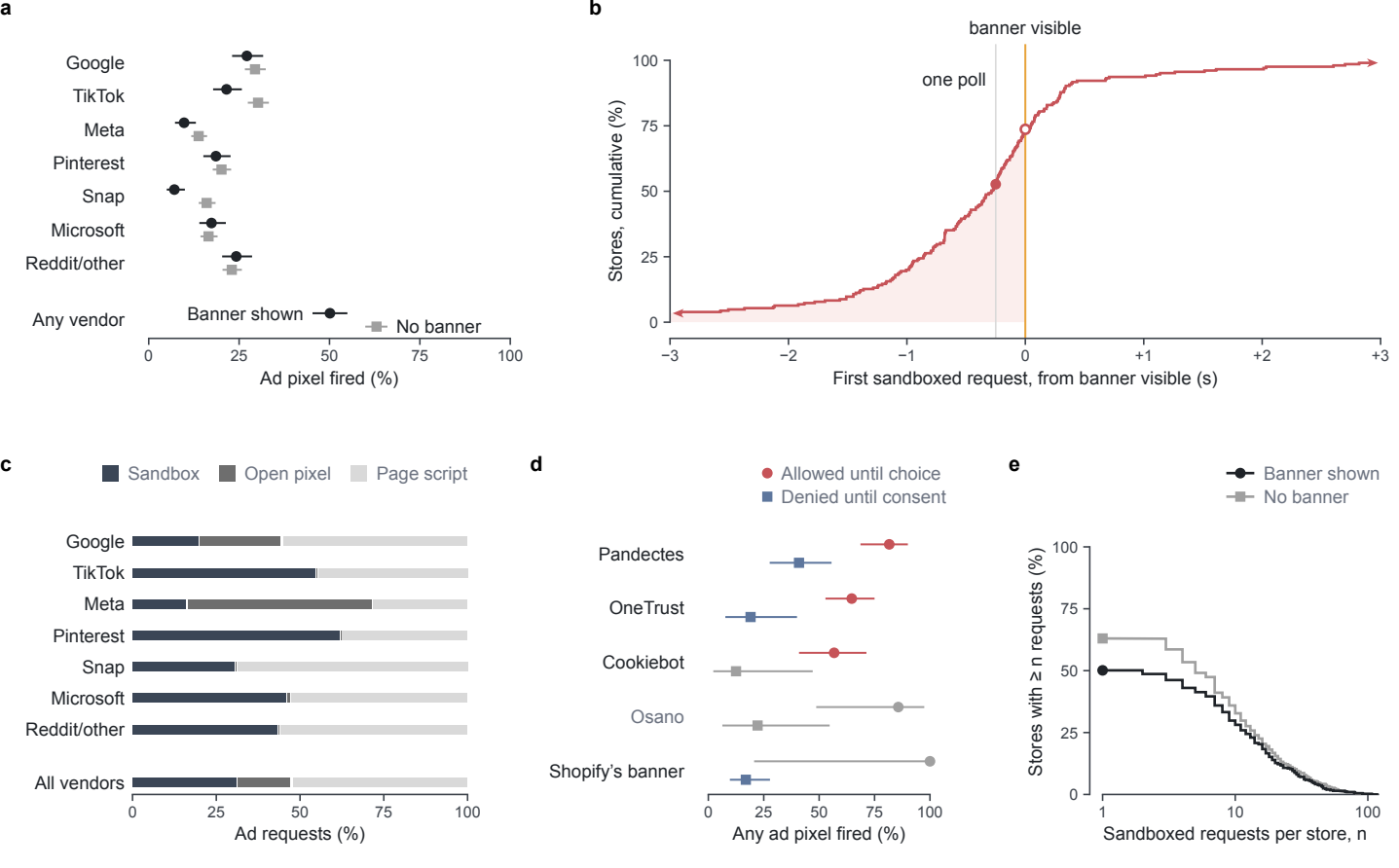

Shopify store in the browser
Page (main thread)
Inline loader
pixel list
Manager bundle
Consent state
cachedConsent
injectedConsent
_cs URL parameter
_tracking_consent cookie
Server-Timing: _cmp
Consent gate
declared purposes
Event bus
page events
CMP's choice
Region default
Global Privacy Control
Pixel sandboxes
Web worker
strict
Sandboxed iframe
lax
Page script
open

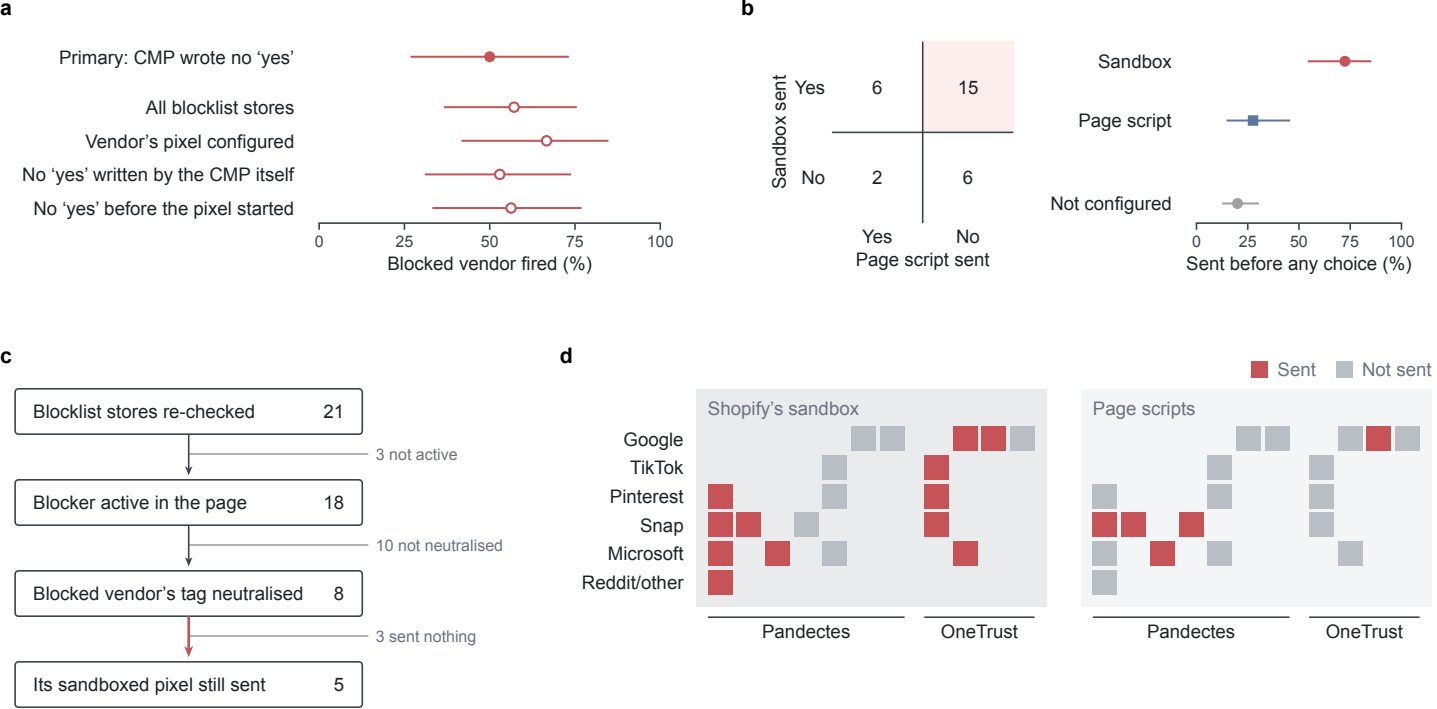

a
Shopify's banner
Writes and blocks
Pandectes
OneTrust
Cookiebot
Osano
Ketch
Clarip
TrustArc
Writes only
iubenda
Blocks only
Usercentrics
Mode unknown
Axeptio
CookieScript
CookieYes
Other or none
0
50
100
Default denies marketing (%)
Fired before any choice (%)
Reject on the first layer (%)
Refusal recorded (%)
Fired on the fresh page (%)
Still fired under GPC (%)
b
Shopify's banner
Pandectes
OneTrust
Cookiebot
Osano
iubenda
All others
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/ other
All vendors
Fired before any choice (%)
100
75
50
25
0

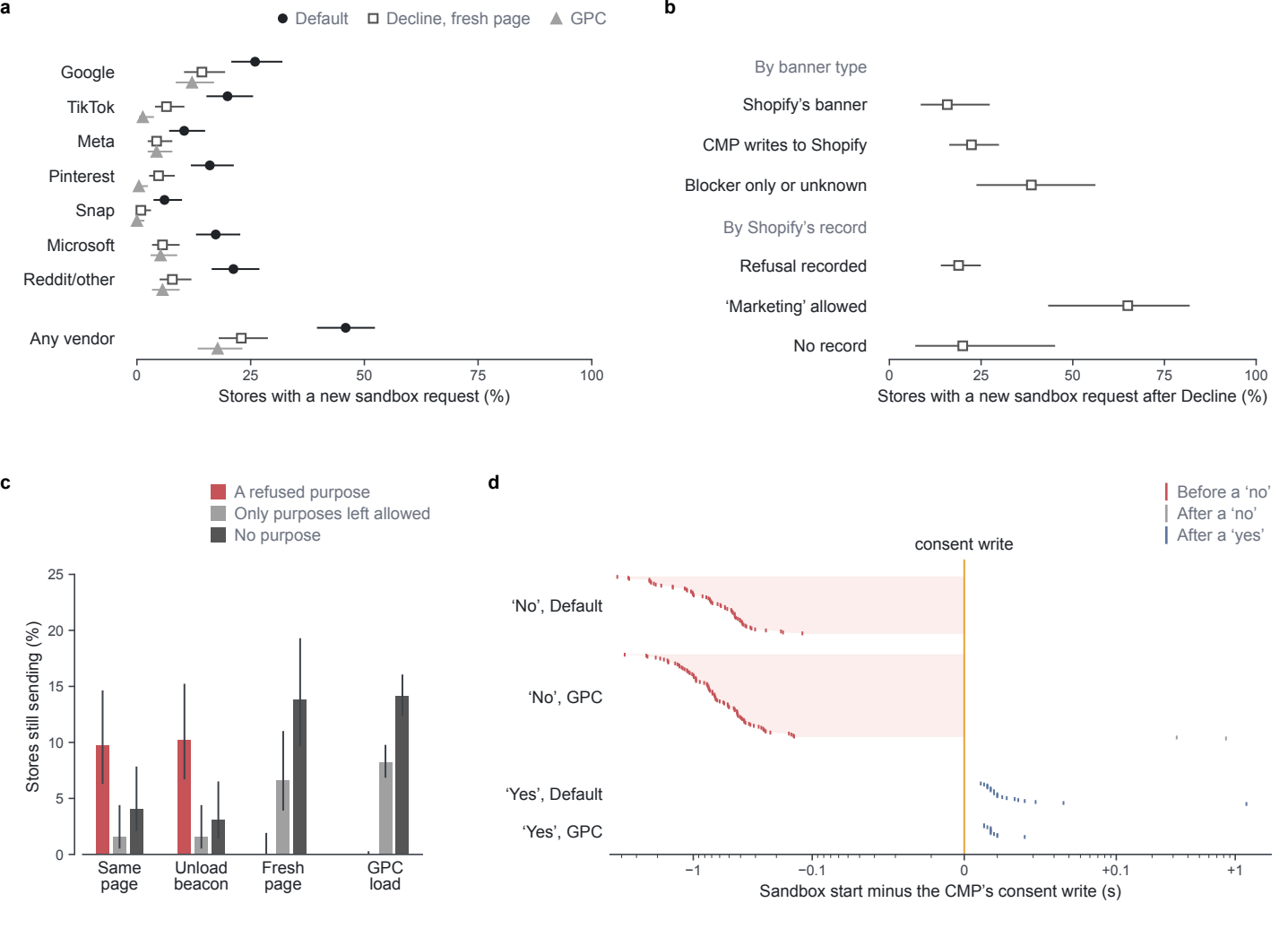

Tranco rank
Top 10k
10k–50k
50k–100k
100k–200k
0
50
100
Banner shown (%)
Fired before any choice (%)
Fired before any choice, banner stores (%)
Fired on the fresh page (%)
Still fired under GPC (%)

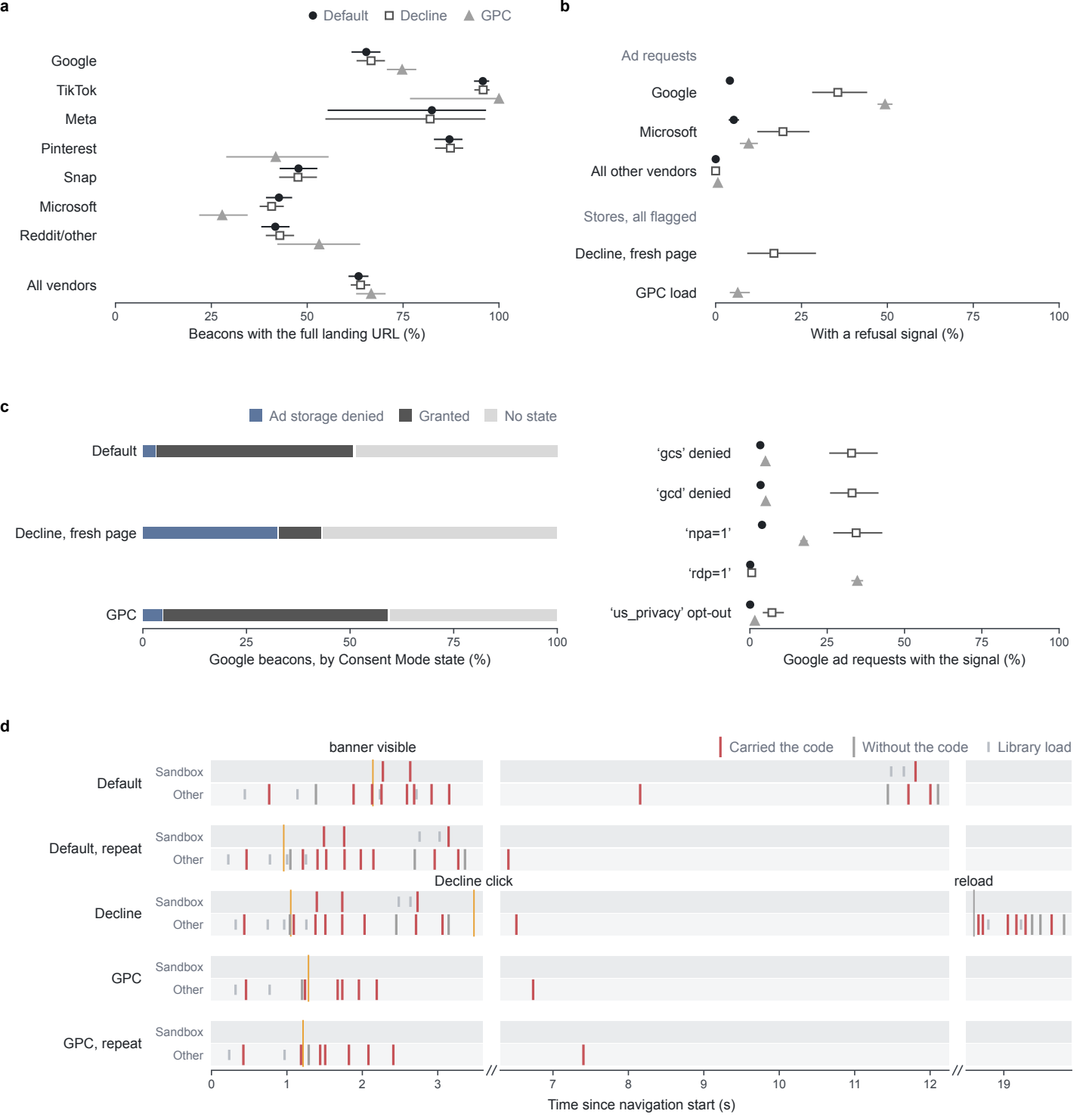

a
Crawler
Yes
No
53
1
not audited
Yes
No
Screenshot audit
b
35
0
1
17
Yes
No
Screenshot audit
c
33
0
0
2
Yes
No
Screenshot audit
d
Cohen's κ
1.00
0.95
0.90
Default vs repeat
GPC vs repeat
GPC vs Brave
Stores
Store × vendor pairs

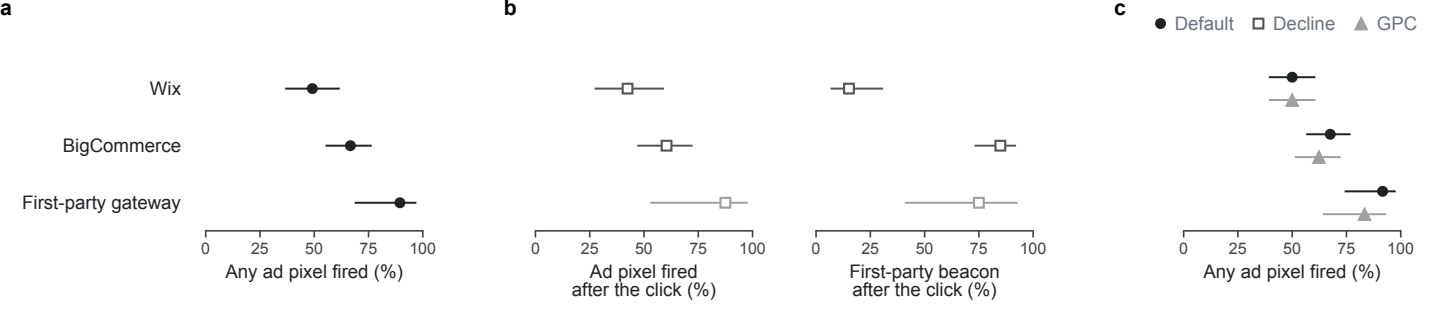

a
Main analysis
Main, bootstrap by consent tool
Crawl segment at 6 browsers
Crawl segment at 10 browsers
Without edge-cached pages
California consent string only
Missed banners added
Consentmo stores as banner stores
Without ethics-flagged stores
Weighted to the Tranco top 200,000
Data beacons only
Beacons with our test code only
Any origin, not only a sandbox
Main analysis
Main, bootstrap interval
Robustness variant
Changed definition
0
25
50
75
100
Sandboxed ad request before any choice (% of stores)
b
Main analysis (fresh page)
Main, bootstrap by consent tool
Including failed forced clicks
Crawl segment at 6 browsers
Crawl segment at 10 browsers
Without any forced click
Without ethics-flagged stores
Weighted to the Tranco top 200,000
Data beacons only
Any time after the click
Fresh page, any sandbox
0
25
50
75
100
Ad request from a new sandbox on the fresh page (% of stores)
c
Main analysis
Main, bootstrap by consent tool
First 10 s of each load
Crawl segment at 6 browsers
Crawl segment at 10 browsers
Without edge-cached pages
Without ethics-flagged stores
Weighted to the Tranco top 200,000
Data beacons only
Beacons with our test code only
Sale-of-data pixels only
Any origin, not only a sandbox
Denominator: all GPC stores
0
25
50
75
100
Sandboxed ad request still sent under GPC (% of stores)


a
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
Any vendor
Banner shown
No banner
0
25
50
75
100
Ad pixel fired (%)
b
banner visible
one poll
Stores, cumulative (%)
100
75
50
25
0
−3
−2
−1
0
+1
+2
+3
First sandboxed request, from banner visible (s)
c
Sandbox
Open pixel
Page script
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
All vendors
0
25
50
75
100
Ad requests (%)
d
Allowed until choice
Denied until consent
Pandectes
OneTrust
Cookiebot
Osano
Shopify's banner
0
25
50
75
100
Any ad pixel fired (%)
e
Banner shown
No banner
Stores with ≥ n requests (%)
100
75
50
25
0
1
10
100
Sandboxed requests per store, n


a
Primary: CMP wrote no 'yes'
All blocklist stores
Vendor's pixel configured
No 'yes' written by the CMP itself
No 'yes' before the pixel started
0
25
50
75
100
Blocked vendor fired (%)
b
Sandbox sent
Yes
No
6
15
2
6
Yes
No
Page script sent
Sandbox
Page script
Not configured
0
25
50
75
100
Sent before any choice (%)
c
Blocklist stores re-checked 21
3 not active
Blocker active in the page 18
10 not neutralised
Blocked vendor's tag neutralised 8
3 sent nothing
Its sandboxed pixel still sent 5
d
Sent
Not sent
Shopify's sandbox
Page scripts
Google
TikTok
Pinterest
Snap
Microsoft
Reddit/other
Pandectes
OneTrust
Pandectes
OneTrust


a
Default
Decline, fresh page
GPC
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
Any vendor
0
25
50
75
100
Stores with a new sandbox request (%)
b
By banner type
Shopify's banner
CMP writes to Shopify
Blocker only or unknown
By Shopify's record
Refusal recorded
'Marketing' allowed
No record
0
25
50
75
100
Stores with a new sandbox request after Decline (%)
c
A refused purpose
Only purposes left allowed
No purpose
Stores still sending (%)
0
5
10
15
20
25
Same page
Unload beacon
Fresh page
GPC load
d
Before a 'no'
After a 'no'
After a 'yes'
consent write
'No', Default
'No', GPC
'Yes', Default
'Yes', GPC
−1
−0.1
0
+0.1
+1
Sandbox start minus the CMP's consent write (s)


a
Default
Decline
GPC
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
All vendors
0
25
50
75
100
Beacons with the full landing URL (%)
b
Ad requests
Google
Microsoft
All other vendors
Stores, all flagged
Decline, fresh page
GPC load
0
25
50
75
100
With a refusal signal (%)
c
Ad storage denied
Granted
No state
Default
Decline, fresh page
GPC
0
25
50
75
100
Google beacons, by Consent Mode state (%)
'gcs' denied
'gcd' denied
'npa=1'
'rdp=1'
'us_privacy' opt-out
0
25
50
75
100
Google ad requests with the signal (%)
d
banner visible
Carried the code
Without the code
Library load
Default
Default, repeat
Decline
GPC
GPC, repeat
Sandbox
Other
Decline click
reload
0
1
2
3
7
8
9
10
11
12
19
Time since navigation start (s)


a
Wix
BigCommerce
First-party gateway
0
25
50
75
100
Any ad pixel fired (%)
b
0
25
50
75
100
Ad pixel fired
after the click (%)
0
25
50
75
100
First-party beacon
after the click (%)
c
Default
Decline
GPC
0
25
50
75
100
Any ad pixel fired (%)